Superblocks for Budapest: automated design, impact assessment and an implementation pathway

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Abstract

A superblock is a group of city blocks, typically 6 to 20 ha, whose interior streets are closed to through motor traffic while every address keeps car access. Through traffic is kept on a surrounding grid of main roads, and the freed interior street space can be given to walking, cycling, play and greenery. We present Superblocker, an open-source tool that designs a superblock plan for a whole city automatically from OpenStreetMap, and apply it to Budapest. The tool chooses the boundary-road network, divides the city into cells, places modal filters, one-way changes, bus gates and camera-enforced destination-only gates, and verifies on the routing graph that no car can cross a cell between two main roads while every junction stays reachable. It then tests the plan in a city-wide traffic, NO_2 and noise model built only from open data. For Budapest the tool produces 2,598 cells in about 100 s. All cells pass the through-traffic test, none of the roughly 4,200 main-road entries can reach another main road through residential streets, and no junction loses car access; the plan needs about 1,300 destination-only gates. The impact model is less favourable. With a modest trip response (elasticity -0.5), average peak car trips take about 21% longer, network capacity falls by about 19%, and population-weighted NO_2 rises slightly (22.4 to 22.8 $\mu g\,m^{-3}$), because diverted traffic concentrates on main roads lined with homes. An 800 HUF entry charge at the Hungária ring calms the centre but moves cross-river traffic to the free ring. Two changes help most: building only the best-ranked quarter of superblocks keeps 78% of the interior traffic reduction at a little over half the delay, and charging the Danube bridges as well closes the ring bypass; together they bring population-weighted NO_2 back to today's level with trips 13% slower. The plan's 1,047 destination-only gates rely on camera enforcement that Hungarian law does not yet allow. We conclude that a city-wide superblock grid is technically constructible for Budapest, but should be built district by district from trial installations, together with measures that reduce car demand, and with measured traffic counts before any number here is quoted as a forecast.

1 Introduction

Most residential streets in a European city carry traffic that has neither its origin nor its destination there. Navigation apps route drivers onto any street that saves a minute, so quiet streets become rat runs, bringing noise, NO_2, crash risk and a carriageway children cannot use.

The superblock (Catalan *superilla*) answers this with network design rather than with speed humps. A cluster of blocks is ringed by main roads that carry through traffic. Inside, a few modal filters, one-way loops or gates make every route between two main roads longer than the route around, while residents, deliveries and emergency services still reach every door. The concept was developed by Salvador Rueda and the Barcelona Urban Ecology Agency [1]; Barcelona's 2016 programme, Ghent's 2017 circulation plan and London's low-traffic neighbourhoods are its best documented applications.

Designing such a plan by hand is slow and error-prone. A filter in the wrong place strands a street, sends traffic through the next block, or blocks a bus route. Most cities therefore build one

superblock at a time, and nobody checks what the whole grid would do. This paper asks three questions for Budapest:

1. Can a complete superblock plan for the whole city be generated automatically from open data, and verified to remove through traffic without cutting off any address?
2. What would such a plan do to car travel times, network capacity, air quality and noise, city-wide and for the people who live on the main roads?
3. What would it take to build it in Budapest: in law, in money, in sequence and in evidence?

Our contributions are an open-source design algorithm with formal validity checks (Section 4), a city-wide screening model using only open data (Sections 3 and 4), results for Budapest with their uncertainties stated (Sections 5 and 6), and an implementation pathway grounded in what worked and failed elsewhere (Section 7).

2 Background and related work

The evidence on superblocks and their relatives is consistent on the inside of a scheme and contested on its edge: inside, traffic falls a lot; on the boundary, outcomes depend on how much traffic disappears rather than diverts.

Barcelona. The Poblenou and Sant Antoni superblocks cut interior traffic sharply; Sant Antoni saw NO_2 fall 25% ($14.6\,\mu g\,m^{-3}$) and PM_{10} 17%, while Horta showed no significant change overall and Poblenou residents reported traffic moving to surrounding streets [6]. A health impact assessment of the full 503-superblock vision estimated about 667 premature deaths avoided each year, mainly from less NO_2, noise and heat and more physical activity [2]. That estimate assumes much of the displaced traffic disappears.

Ghent. In 2017 Ghent divided its centre into six sectors that cars can only move between via the ring. Peak car traffic fell about 12%, cycling rose about 25% and public transport about 28%. Ghent is the closest precedent to a whole-city design.

London low-traffic neighbourhoods. Across 46 London LTNs, mean interior traffic fell from 1,780 to 930 vehicles against 1,745 expected from background trends, while boundary roads changed only marginally on average [5]. Results vary widely between schemes, and boundary-road congestion is the most common public objection.

Leuven and Vienna. Leuven's 2016 circulation plan cut car traffic in the centre by 8% and raised cycling by 32%, while car traffic on the ring rose 9% and stayed within capacity [7]. A MATSim study of Vienna simulated 61 configurations with up to 46 superblocks and found interior car trips fall roughly linearly, about 100 a day per superblock [8].

Traffic evaporation. A review of about 100 road-space reductions found traffic on the treated roads fell 41% on average, and less than half of it reappeared on nearby roads [4]. Evaporation is real but not guaranteed; it depends on alternatives existing.

Road pricing. Stockholm's cordon cut entries by 20–22% and held at 18–20%; Milan's Area C by 30–38%; London's area charge by 18%; Gothenburg's by 12%, with its bypasses 16–26% busier [14–18]. Pricing reduces demand, whereas filters only redistribute it. London's experience is that taking road space away without pricing lets congestion return while volumes stay down.

Automated design. Eggimann [3] identified candidate superblocks in many cities from street-network geometry, showing the potential is large and widespread. To our knowledge no published tool designs the filter placement for a whole city, verifies the result free of through routes and stranded addresses, and tests it in a city-wide traffic assignment. That is the gap this paper addresses.

Budapest. Budapest has district-level traffic calming and some *kivéve célforgalom* (access only) streets, but no superblock programme. An inner-city entry charge has been studied since 2008 [22] and is in the 2019 Mobility Plan [19], but Hungarian road law does not let the capital levy it [20] (Section 7).

3 Data

Every input is open and needs no account, so the study can be repeated by anyone and for any city (Table 1). Traffic counts are the exception and the study's largest weakness. They exist: several hundred Budapest Közút loop and camera counters report daily volumes since October 2022, and BKK's strategic model EFM 3.0 (PTV VISUM, over 1,000 zones, rebuilt in 2024 from a household survey of about 60,000 trips) is released to researchers on request [9, 10]. Neither is published as open files, and neither was used here.

Table 1: Data layers.

Layer	Source	Licence	Use
Street network (34,641 junctions)	OpenStreetMap via OSMnx [23], inside the Nominatim boundary	ODbL	Cells, filters, routing, capacities
Turn restrictions	OSM relations via Overture Maps	ODbL	Movement-level routing
Bus, trolleybus, tram routes	OSM; BKK GTFS feed	ODbL; BKK	Bus gates; walking access to stops
Fire, ambulance, schools	OSM; Overture places	ODbL; CDLA	Emergency response; school streets
Residents	Meta High Resolution Settlement Layer, 30 m	CC BY 4.0	Trip origins; exposure weighting
Jobs (proxy)	Overture places and buildings	CDLA, ODbL	Trip destinations
NO_2 annual means 2025	10 Budapest stations via OpenAQ	CC BY 4.0	Fitting the NO_2 model
Entry-charge cordon	Hungária ring and Buda inner ring (32 km^2)	ODbL	Entry charge scenario
Traffic counts	Budapest Közút counters (dashboard only); EFM 3.0 on request	–	Not used; would calibrate demand and boundary roads
Peak congestion	TomTom Traffic Index 2025 [11]	public	Range for the calibration target

The software is written in Python 3.12 (FastAPI, OSMnx, NetworkX, GeoPandas, Shapely) with a React, MapLibre and deck.gl interface, and is released under GPL-3.0 at `https://github.com/tgergo1/Superblocker`.

4 Method

The method has two halves (Figure 1). The design half builds a plan and verifies two properties on the routing graph: no through route, and no address cut off. The assessment half asks what the plan does to the city in a traffic, air and noise model.

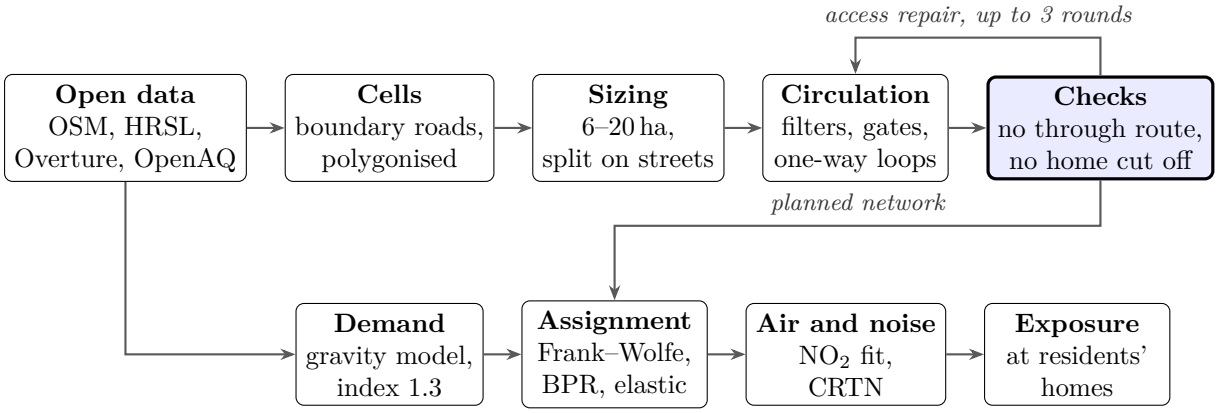


Figure 1: Design and assessment pipeline. The top row designs the plan and loops through the access repair until every check passes; only a valid plan enters the city-wide traffic, air and noise model in the bottom row.

4.1 Boundary roads and cells

The street graph is loaded inside the city's administrative polygon. Boundary roads are chosen from measured counts when they exist; otherwise from road class (motorway to secondary, plus busy tertiary roads) and sampled betweenness centrality. The boundary network is polygonised into closed cells and clipped to the city boundary. Motorways and trunk roads are corridors the plan never modifies.

4.2 Sizing

Cells are merged and split towards a target of 12 ha, with 6 ha and 20 ha as the limits. An oversized cell is split along the whole length of a residential street, followed through its junctions to the boundary roads. Among candidate splits, the one chosen divides the area most evenly while keeping its least compact piece compact, measured by the Polsby–Popper score [27]

$$PP = \frac{4\pi A}{P^2}, \tag{1}$$

where A is the piece's area and P its perimeter. A split street gets a filter halfway along so that it does not become a new rat run. Housing estates served only by living streets may split along them as a last resort. Three kinds of cell are reported separately rather than forced into range: cells already calm (all interior streets are dead ends from one entry), cells that are mostly open land, and undividable blocks where every split leaves a sliver.

4.3 Circulation design

Every junction where a cell's streets meet a boundary road is an entry. Entries are grouped into sectors (four by default, falling back to three and then two when a design cannot keep access). A constraint solver chooses the cheapest set of changes so that no directed path inside the cell links two different sectors; the changes, in rising cost, are a one-way conversion, a modal filter and a full street cut. A filter on a bus, trolleybus or tram route becomes a camera bus gate; other filters are emergency bollards. Where every cut that would close a route also strands homes, the closure becomes a camera-enforced destination-only gate that residents, deliveries and visitors may pass.

4.4 Validity checks

A plan is accepted only when it passes three checks on the routing graph. Through traffic is routed as an ordinary car; access is routed as a resident who may pass destination-only gates.

1. **No through route.** For each cell, no car path enters at one sector and leaves at another.
2. **No crossing.** No car entering from a main road can reach another main-road junction through residential streets, including across neighbouring cells. A seal pass adds filters (up to 30 per entry) until this holds.
3. **No stranded junction.** On the whole planned city, every junction reachable by car today stays reachable in and out. An access repair reopens the cheapest change for each stranded pocket (a one-way first, then a split-street filter, then an interior filter), turns it into a gate when reopening would recreate a through route, and redesigns the affected cells for up to three rounds.

4.5 Traffic model

The city is divided into 800 m zones. Peak-hour car trips run between zones by a gravity model, from residents to jobs, with 15% of trips entering or leaving through main-road gateways at the city edge. Demand is scaled so that today's average trip takes 1.3 times its free-flow time. Trips are assigned to user equilibrium by the Frank–Wolfe algorithm [24], movement by movement (turn restrictions, junction delays by control type, U-turns only at dead ends), with BPR link costs [25]

$$t(v) = t_0 \left[1 + 0.15 \left(\frac{v}{c} \right)^4 \right]. \tag{2}$$

Demand then responds to the plan with an elasticity of -0.5 on trip time, iterated with the assignment. Capacity is the demand multiple at which the average trip takes twice its free-flow time.

4.6 Air and noise

Each link emits NO_x in proportion to its volume and a speed-dependent fleet emission factor, with a heavy-vehicle share by road type. Emissions are spread on a metric grid by a near-road kernel (exponential, 30 m) and an urban-scale kernel (4 km). Annual mean NO_2 at the ten stations is fitted by non-negative least squares,

$$\mathrm{NO_2} = b_0 + a\, E_{\mathrm{near}} + b\, E_{\mathrm{urban}}, \tag{3}$$

with the two decay lengths chosen by leave-one-out error. Noise is the CRTN peak-hour level at 10 m from each link [26], spread by inverse square and summed by energy, without screening by buildings. Both are counted at residents' homes.

4.7 Entry charge

A charge at the Hungária-ring cordon enters route choice as time at a value of time of 3,000 HUF/h, so through traffic can go round. A trip whose route still enters the zone keeps a share of its demand

$$s = e + (1 - e) \left(\frac{C + p}{C} \right)^{\varepsilon}, \tag{4}$$

where C is the trip's generalised cost (time at 3,000 HUF/h plus 60 HUF/km), p the charge (800 HUF central), $e = 0.2$ the exempt share and $\varepsilon = -0.7$ (range -0.5 to -1.2, from Stockholm and Gothenburg at introduction [16, 17]). Zone residents are exempt. Junctions within 60 m of the cordon count as outside, so the ring and the end bridges are free.

5 Results

The tool can build a valid superblock plan for all of Budapest, but the plan on its own makes driving slower and does not improve air quality on average. Figures come from three runs of the software at successive versions (October 2026); each table names its run. Cell counts differ between runs because the impact runs use the population layer, which changes which cells count as superblocks.

5.1 The plan

Table 2 gives the whole-city plan at default settings without traffic counts.

Table 2: Whole-Budapest plan (12 ha target, 6–20 ha limits).

Result	Value
Cells	2,598
Cells passing the through-route check	2,598 (100%)
Main-road entries reaching another main road via residential streets	0 of ≈4,200
Junctions cut off from car access	0
Destination-only camera gates	≈1,308
Undividable blocks over 20 ha (median 24 ha)	43
Cells over 20 ha that are mostly open land (median 48 ha)	120
Superblocks under 6 ha	5
Eligible road coverage of the boundary network	99.8%
Run time	≈100 s

The destination-only gates are the price of validity. About 930 of them sit on split streets the access repair had to reopen, 760 of those on a single block-long section with no other place for a filter. A plan with only physical filters would strand homes or leave through routes.

5.2 Traffic, air and noise

Table 3 gives impact model v2 (2,313 superblocks, 646 bus gates, no motorway changes, 238,000 morning-peak car trips).

Table 3: Traffic impact of the plan against today.

Measure	Fixed demand	Trips respond ($\varepsilon = -0.5$)
Average car trip time	+21.5%	+20.6%
Car-km	+9.7%	+4.0%
Peak car trips not made by car	0	8.7%
Network capacity	−19%	−19%
Junctions cut off once filtered sections close	11	11

The NO_2 fit reproduces the ten stations with $R^2 = 0.59$ and a leave-one-out error of $5.0\,\mu g\,m^{-3}$. Population-weighted annual NO_2 goes from 22.4 to $22.8\,\mu g\,m^{-3}$: 169,000 residents get at least $1\,\mu g\,m^{-3}$ less and 307,000 at least 1 more. For noise, 281,000 residents get homes at least 3 dB quieter and 240,000 at least 3 dB louder.

The distribution matters more than the averages. Interior streets get much quieter and cleaner, but Budapest's main roads are lined with flats, so the people who take the diverted traffic are many. Without demand falling, the plan moves exposure rather than reducing it.

5.3 Entry charge

Table 4 compares the plan, an 800 HUF charge per entry at the Hungária-ring cordon (32 km^2, 108 charged entry links) and both together (205,825 morning-peak car trips, 2,123 superblocks, elasticity −0.7 with the range in brackets).

Table 4: Entry charge scenarios against today, once trips respond.

Measure	Superblocks	Charge	Both
Cars entering the zone (57,600/h today)	+11%	−76% (−75 to −79)	−68% (−66 to −71)
Driving inside the zone	+9%	−44%	−34%
Main roads within 400 m outside the zone	+49%	+80%	+111%
Average car trip time	+19.4%	+7.9%	+25.8%
Main roads more than 10% busier	773 km	335 km	751 km
Peak car trips not made	8.1%	3.1%	11.9%
Residents ≥1 µg m^{-3} more / less NO_2	342k / 36k	61k / 64k	230k / 63k
Residents ≥3 dB louder / quieter	211k / 308k	83k / 113k	222k / 354k
Revenue per year	–	18 bn HUF	25 bn HUF

The charge calms the centre and, with the plan, cuts the residents breathing worse air from 342,000 to 230,000. It does not relieve the main roads the plan loads (773 km against 751 km), because cross-river traffic leaves the central bridges for the free ring and end bridges; peak flow on the Rákóczi bridge rises from about 5,200 to 9,100 vehicles an hour with the charge alone. At 400 HUF the charge alone still cuts entries by 59%: most of the effect is rerouting, not trips given up.

5.4 Sensitivity and improvements

A November 2026 run on the current code tests the weakest assumptions and three ways to improve the plan. It rebuilds the plan with the population layer (2,291 superblocks, all valid, 1,047 destination-only gates) and reproduces the impact model within a few points: 198,000 peak car trips, average trip time +18.2% with today's trips and +18.9% once 8.3% of trips are not made by car, capacity −17%, population-weighted NO_2 22.4 to 23.6 µg m^{-3} (23.1 with the trip response). Table 5 summarises the variants against this run.

Table 5: Variants of the November 2026 run (trips respond, $\varepsilon = -0.5$, except where noted). Trip time and NO_2 are against today in the same model.

Variant	Superblocks	Trip time	Capacity	NO_2 (µg m^{-3})
Full plan, congestion index 1.3 (as in the paper)	2,291	+18.9%	−17%	23.05
Full plan, index 1.6 (TomTom all-day)	2,291	+20.7%	−12%[a]	22.99
Full plan, index 1.9 (TomTom morning peak)	2,291	+24.4%	−10%[a]	22.96
Full plan, gates not enforced	2,291	+13.6%	−14%	22.88
Best 75% of superblocks	1,718	+16.5%	−14%	22.99
Best 50%	1,146	+16.1%	−12%	22.98
Best 25%	573	+11.4%	−7%	22.88
Best 10%	229	+4.5%	−1%	22.62
Charge at the Hungária cordon, 800 HUF[b]	2,291	+25.3%	–	22.97
Same charge, Danube bridges charged too[b]	2,291	+20.9%	–	22.61
Danube bridge charge without superblocks[b]	0	+3.2%	–	22.19
Best 25% with the Danube bridges charged[b]	573	+13.3%	–	22.45

[a] Capacity is the demand at which trips take 1.54 times today's level in each case, so the three are not directly comparable. [b] Charge elasticity −0.7. Today's NO_2 is 22.42 (index 1.3), 22.35 (1.6) and 22.27 (1.9).

Congestion level. The model assumes today's peak trip takes 1.3 times its free-flow time. TomTom measured 61.9% congestion across 2025 and 88.9% in the morning peak, about 28 and 33 minutes per 10 km [11], which suggests an index between 1.6 and 1.9. TomTom's figure is distance-weighted over probe vehicles, not the model's trip-weighted ratio, so we treat it as a range. Calibrated to it, demand rises to 258,000 and 295,000 trips and the plan's trip-time penalty grows from +19% to +21% and +24%. The direction of every result holds; the penalty in the paper is, if anything, understated.

Unenforced gates. Camera enforcement is not legally available today (Section 7). With gates that cost through traffic nothing, the trip-time penalty falls from +19% to +14%, because drivers cut through the gated streets the plan meant to protect. Traffic on the gated links rises from about 3,800 to 14,900 vehicle-km in the peak hour, and the interior traffic the plan removes falls by 8%. The gates are therefore load-bearing: without enforcement part of the plan's effect, good and bad, leaks back onto residential streets, so the gates that carry most of this traffic should become rising bollards with resident access until camera enforcement is allowed.

Coverage frontier. Ranking superblocks by interior traffic removed per unit of extra boundary-road traffic (both from the full-plan assignment; Section 4) shows that the benefit is concentrated. In the full plan, 1,366 of 2,291 superblocks (60%) carry no modelled through traffic today, so filtering them changes nothing inside and only lengthens access trips. The best 10% of superblocks remove 39% of the interior vehicle-km the full plan removes, at a seventh of its trip-time penalty (a quarter once trips respond); the best 25% remove 78% at a little over half of the penalty; beyond half, more superblocks add delay and almost no interior benefit (Table 6). The interior benefit saturates after the first quarter while delay keeps rising, because the remaining superblocks have little through traffic to remove but still lengthen access trips and load the boundary roads. The model's 800 m zones load few trips onto quiet interiors, so the frontier understates the local benefit of calming streets that are already quiet; the ranking is still the right order in which to build.

Table 6: Coverage frontier: share of the full plan's interior benefit against its costs (fixed demand; trip response in brackets).

Built	Interior veh-km removed	Trip time	Capacity	Boundary veh-km added
10% (229)	39%	+2.5% (+4.5%)	−1%	26%
25% (573)	78%	+9.9% (+11.4%)	−7%	73%
50% (1,146)	92%	+14.0% (+16.1%)	−12%	89%
75% (1,718)	92%	+14.8% (+16.5%)	−14%	91%
100% (2,291)	100%	+18.2% (+18.9%)	−17%	100%

Charging the bridges. The Hungária cordon failed because cross-river traffic moved to the free ring and end bridges. Charging every Danube road bridge as well (800 HUF per crossing, residents of the zone exempt) closes that bypass: with the plan, trip time rises 21% instead of 25%, ring-road traffic 78% instead of 112%, and population-weighted NO_2 is 22.6 instead of 23.0 µg m^{-3}. On its own the bridge charge lowers NO_2 to 22.2 at a 3% trip-time cost, the only variant that improves air quality on average. The modelled revenue, about 63 bn HUF a year, reflects charging every crossing at a cordon price; a real scheme would charge less per crossing or once a day. Combining the two improvements, the best 25% of superblocks with the bridges charged keeps 78% of the interior benefit with trips 13% slower and population-weighted NO_2 back at today's level (22.45 against 22.42 µg m^{-3}), against +25% and 23.0 for the full plan with the cordon charge.

6 Limitations and validity

The design checks are verifications on a graph; the impact numbers are screening estimates. The two should not be read with the same confidence.

What is verified, and on what. The no-through-route, no-crossing and no-stranded-junction properties hold on the OpenStreetMap graph as mapped. They say nothing about map errors (a missing alley, a wrong one-way tag), private access roads, or addresses not snapped to the graph. The tool accepts an authoritative address dataset and checks each target, but none was used here.

No traffic counts. Budapest publishes no open link counts. Demand is a gravity model calibrated to a single travel-time index of 1.3, and boundary roads come from road class and centrality. The modelled Hungária körút peaks at about 3,000 vehicles an hour, probably below reality. Every traffic figure is therefore relative (plan against today in the same model), not a forecast.

Static, single-class assignment. Every driver takes the cheapest route at one value of time. Real drivers differ, and many would pay a charge or accept a slower route. This is why the modelled entry cut of -68 to -79% is far beyond the -12 to -30% measured elsewhere, and should be read as an upper bound.

Evaporation is probably understated. The trip response gives 8.7% fewer peak car trips. Measured schemes saw larger falls on treated roads (41% on average [4]), helped by better transit and cycling, which the model does not add. The trip-time and NO_2 penalties are closer to a pessimistic case than a central one.

Air and noise. The NO_2 fit explains 59% of the variance at only ten stations, treats chemistry as linear and uses a fleet-average emission factor. Noise ignores screening by buildings, so back façades are too loud in absolute terms. Changes are more robust than levels.

Not modelled. Public transport and cycling demand, parking, freight timing, compliance at gates, construction cost beyond indicative unit costs, and long-run land use. Each would move the results, mostly in the plan's favour if the city invests in alternatives at the same time.

7 Implementing superblocks in Budapest

Budapest should build superblocks as Barcelona and London did, a few at a time from reversible trials, and add demand management before the grid is dense enough to load the main roads. The city-wide plan is the map that keeps each local scheme consistent with the next, not a single project.

7.1 Who decides

Budapest has two tiers of government. The capital and its transport authority BKK manage the main-road network and public transport; the 23 district councils manage most residential streets. Superblock interiors are therefore district decisions, while boundary roads, bus gates and any charge are the capital's. Every scheme needs a joint agreement between a district and BKK, with BKK checking boundary-road capacity and bus routes.

7.2 Legal instruments

None of the plan's street measures needs new legislation: modal filters, one-way changes, 30 km/h zones and the "Behajtani tilos, kivéve célforgalom" sign are ordinary traffic orders under the KRESZ. Camera enforcement is the plan's main legal risk. Municipal police may only fine in person, and in April 2025 the Interior Ministry refused the capital's request to let number-plate cameras fine drivers in bus lanes [12]. Until that changes, a destination-only gate is enforced only by occasional police checks; Section 5.4 measures what that costs, and the gates that matter should become rising bollards with resident access. The entry charge is different: the road traffic act (Kkt.) reserves road charges to the state, and the Kúria annulled the capital's lorry fee on that ground in 2024 [20]. A charge needs Parliament to amend the Kkt. first [21].

7.3 Phasing

1. **Evidence (months 0–6).** Request the EFM 3.0 matrices and the counters' hourly data from BKK. Install counters on the boundary roads and candidate interiors of the first districts (Telraam sensors, BKK counts, published bridge counts). Load an authoritative address dataset. Rerun the plan with counts so the boundary network comes from measured volumes.
2. **Pilots (months 6–18).** Pick three to five superblocks from the tool's pilot ranking, which favours large interior traffic cuts, dense housing, schools and short detours. Build them with planters, removable bollards and signs; in an earlier run the ten top-ranked trials cost about €160,000 in total. Measure inside and on the boundary before and after, and publish the results.
3. **District rollout (years 2–5).** Extend to whole districts where pilots work, starting where tram, metro and cycling alternatives are best (inner Pest districts VI–IX and XIII). Give the reclaimed space to people: greenery, seating, school streets, delivery bays. Pair each district with bus priority and cycle routes on its boundary roads.
4. **City grid and demand management (year 3 onwards).** Before the grid covers most of the city, introduce parking pricing and supply limits (within the capital's and districts' powers today; hourly rates rose to 800, 600, 400 and 300 HUF in zones A to D on 1 July 2026 [13]), then a charge once the law allows. The model shows the charge must keep cross-river through traffic charged (end bridges too, or a per-day area charge as in London and Milan), or it simply moves traffic onto the ring.

7.4 Gates between phases

The tool's release gate already refuses "implementation ready" until measured counts cover at least 80% of road length, the address dataset is complete, and engineering and site reviews are recorded. We propose the city adopts the same rule. In addition, a phase should proceed only if its measured results meet targets agreed in advance: interior traffic down by at least a third, no boundary road with sustained queues beyond an agreed limit, bus journey times not worse, and emergency response within one minute of today.

7.5 Monitoring and rollback

Each scheme is installed with counts before, three months after and a year after, and with a pre-agreed rollback rule; trial materials make rollback cheap. Ghent and London kept schemes through difficult first months because measurements, not anecdotes, led the debate. The failed charges in Edinburgh and Manchester show the opposite risk: benefits not visible before the vote, and revenue use not trusted.

7.6 Cost and funding

Indicative central European unit costs in an earlier run gave about €7.6M for trial installations across the whole city and about €24.8M for permanent works, before public-space redesign; these are screening figures to be replaced by engineering estimates. A charge at 800 HUF would raise an estimated 18–25 bn HUF a year, enough to fund the programme many times over.

8 Conclusion

A complete superblock plan for Budapest can be generated from open data in under two minutes and verified free of through routes and stranded addresses on the street graph. The design problem is solved well enough to serve as a planning instrument. What remains is the city problem: the same model says that redistributing traffic without reducing it lengthens trips by about a fifth and moves exposure onto main-road residents.

The sensitivity runs sharpen this. The interior benefit is concentrated: 60% of the superblocks carry no modelled through traffic, and the best quarter delivers 78% of the benefit. So the city-wide plan should be read as a ranked framework rather than a single project: build the part of the grid the boundary roads can carry, in rank order, and grow it as demand falls. The practical answer is to build it district by district from measured pilots, invest in transit and cycling on the boundary roads at the same time, and add demand management that cannot be bypassed on the ring, such as a charge that includes the Danube bridges. Gates that carry the most leaked traffic should become rising bollards until camera enforcement is legal. The next scientific step is calibration: BKK's EFM 3.0 matrices and hourly counter data, a mode-choice step using the GTFS feed, a multi-class assignment with a spread of values of time, and a per-day charge design.

Data and code availability

Code, tests and research notes are at `https://github.com/tgergo1/Superblocker` (GPL-3.0). The traffic, air and noise results come from `POST /api/v1/simulation/capacity` with the layers downloaded by `backend/scripts/fetch_open_data.py`. Entry-charge calibration is in `docs/research/congestion-charging.md`, the cordon in `docs/research/cordon-hungaria.json`, and the design history in `docs/roadmap-budapest.md`. Section 5 figures come from commits `bbca81d` (impact model) and `3b4d346` (entry charge); Section 5.4 from `backend/scripts/run_budapest.py` on the November 2026 code, with the run settings listed in `docs/research/november-2026-runs.md`.

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